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Project      : Sales Analysis using SQL
File Name    : 02_Star_Schema.sql
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Description :
This script creates the Star Schema for the Sales Analysis project.
It includes all dimension tables and the central fact table along
with primary and foreign key relationships.

Tables Created:
1. dim_product
2. dim_date
3. dim_region
4. dim_customer
5. dim_sales_rep
6. dim_sales_channel
7. dim_payment
8. fact_sales

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=====
-- STEP 1: Create Product Dimension
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CREATE TABLE dim_product (
    product_key INT PRIMARY KEY AUTO_INCREMENT,
    product_category VARCHAR(50) NOT NULL
);

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=====
-- STEP 2: Create Date Dimension
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CREATE TABLE dim_date (
    date_key INT PRIMARY KEY AUTO_INCREMENT,
    full_date DATE NOT NULL,
    year INT,
    month INT,
    quarter INT,
    day INT,
    month_name VARCHAR(10),

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        day_name VARCHAR(10)
    );

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=====
-- STEP 3: Create Region Dimension
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CREATE TABLE dim_region (
    region_key INT PRIMARY KEY AUTO_INCREMENT,
    region VARCHAR(50)
);

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-- STEP 4: Create Customer Dimension
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CREATE TABLE dim_customer (
    customer_key INT PRIMARY KEY AUTO_INCREMENT,
    customer_type VARCHAR(50) NOT NULL
);

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-- STEP 5: Create Sales Representative Dimension
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CREATE TABLE dim_sales_rep (
    sales_rep_key INT PRIMARY KEY AUTO_INCREMENT,
    sales_rep VARCHAR(50)
);

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-- STEP 6: Create Sales Channel Dimension
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CREATE TABLE dim_sales_channel (
    sales_channel_key INT PRIMARY KEY AUTO_INCREMENT,
    sales_channel VARCHAR(50) NOT NULL
);

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-- STEP 7: Create Payment Dimension
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CREATE TABLE dim_payment (
    payment_key INT PRIMARY KEY AUTO_INCREMENT,
    payment_method VARCHAR(50) NOT NULL
);

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=====
-- STEP 8: Create Fact Sales Table
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CREATE TABLE fact_sales (

    sales_key INT PRIMARY KEY AUTO_INCREMENT,

    product_key INT,
    date_key INT,
    region_key INT,
    customer_key INT,
    sales_rep_key INT,
    sales_channel_key INT,
    payment_key INT,

    sales_amount DECIMAL(10,2),
    quantity_sold INT,
    unit_cost DECIMAL(10,2),
    unit_price DECIMAL(10,2),
    discount DECIMAL(10,2),

    FOREIGN KEY (product_key) REFERENCES dim_product(product_key),
    FOREIGN KEY (date_key) REFERENCES dim_date(date_key),
    FOREIGN KEY (region_key) REFERENCES dim_region(region_key),
    FOREIGN KEY (customer_key) REFERENCES dim_customer(customer_key),
    FOREIGN KEY (sales_rep_key) REFERENCES dim_sales_rep(sales_rep_key),
    FOREIGN KEY (sales_channel_key) REFERENCES
dim_sales_channel(sales_channel_key),
    FOREIGN KEY (payment_key) REFERENCES dim_payment(payment_key)

);

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